World of Zoo

Wii to PC conversion: complete technical documentation

Everything reverse engineered, built, and tested in the project, as of **24 October 2026**

Scope	Package container, property files, textures, appearance/color system, save file, Havok models and physics, language packages, the tool chain and its options, the complete in-game test history, known bugs, optional Penguin-only fixes.
This document	Updated an earlier summary to correct mistakes and new discoveries. **C**orrections are listed in section 14.
Evidence legend	**[verified]** checked against real Wii / PC file pairs, by byte comparison, or by an in-game test. **[observed]** read directly from files in this project but not compared against a second source. **[assumed]** reasoned, not proven. **[unknown]** not investigated, or investigated without result.
Working style	Claude deeply analyzed the file's content and wrote Python tools. User ran the game and reported crashes and visuals. Tests were built as small packages that isolated one change. The user had the final decision on what went into the tools.

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1. Project goal and current status

Goal. Bring animals that exist only in the Wii release of World of Zoo (Bear, Penguin, Crocodile, Antelopes, Elephants) into the PC retail game (`WoZRetail.exe`, a 32-bit x86 Windows build, 7,050,552 bytes). The animals were already restored on the title screen; the user edited **StartPC.cfg** and placed original Wii files in the PC game directory.
Each animal needs its own animal, editor, habitat and nursery packages, so all of them must be converted.

Core finding [verified]. Wii and PC use the same `.pkg` container and mostly the same data. The Wii files are Big-Endian and use Wii-specific texture and vertex formats. Nothing is encrypted: every file inside a package is plain zlib. Conversion is therefore almost entirely a byte-order and format job, plus a few format-specific rules that the PC engine is strict about (section 8 and section 10).

1.1 Status per animal (user-reported, in game)

Animal	Packages	State
Penguin	PenguinAnimal, Preloading, Editor, StarAnimals, Templates, PenguinPool (+Icons, +Psys), NurseryPool (+Icons, +Psys)	Whole set loads and plays. Colors and patterns work. Two **optional** Penguin-only fixes exist (hold point, nursery collision). Residual: held items look too wobbly when carried by the animal (not a conversion bug).
Bear	BearAnimal, Preloading, Editor, BearForest (+Icons, +Psys), NurseryBear (+Icons, +Psys)	Works: model, enclosure, editor colors and patterns. Swapping enclosure decorations with the hammer had to be fixed due to a mistake with the cylinder-shape hitbox of some models.
Crocodile	CrocodileAnimal, Preloading, NurseryPaddockCrocodile	Converted and working. The nursery package used to fail on `scripts/settingmodebase.dsc`; fixed (section 6.4). Repeated item requests were researched as a bug; reconsidered to be original Wii behavior.
Gazelle	GazelleSavanna (habitat)	Converted and working; hammer tool works.
Big Cat	11 native PC .pkg files	Reference only. Its Wii and PC packages form the ground-truth pairs used to learn every format.

Net result: all converted animals work as intended. What remains are small, non-blocking issues listed in section 13, plus the decision about which optional Penguin fixes become part of the tools.

2. Terminology

- **Patterns** = the selectable appearance options in the game. **Textures** = the underlying image/layer files.
- The Animal Editor has **six color controls**: Main Color (primary body color), Second to Fifth Color (pattern-dependent regions; the target differs per animal and pattern, e.g. Horse hooves, Big Cat nose), Sixth Color (usually the eyes).
- A **dirt layer** is applied on top of the generated appearance while the animal is in its enclosure or nursery.
- Editor part lists: **Bear** = Head, Nose, Chin, Ears, Body, Paws, Tail. **Big Cat** = Head, Nose, Mouth, Ears, Whiskers, Body, Paws, Tail. Rest of the animals differ in other ways.
- "Preloading", "Editor", "StarAnimals", "Templates" are package suffixes. The "Psys" and "Icons" suffixes hold particle systems and UI icons for a habitat or nursery.

3. Tool chain

All tools are Python 3, standard library only (Pillow is used only by throw-away analysis scripts, never by the tools). `wozconvert.py` **is the main converter script**: it embeds **`wozpkg`, `wozdsc`, `wozhkx`,** and **`wozfix`** as compressed blobs, so it runs standalone. The separate scripts are exports of the same code for individual use. The embedded modules are always the newest; though as of the current date, all individual scripts have been updated to match `wozconvert.py` modules.

File	Purpose	Notes
`wozconvert.py`	All-in-one converter: `python wozconvert.py WiiFolderOrPkg OutFolderOrPkg [options]`. Converts every file type it knows, rewrites the container in PC format, prints a report. Lines "STILL WII FORMAT" or "FAILED" need attention.	Detailed usage of the converter is included in a commentary at the first lines of the tool itself.
`wozpkg.py`	Container library CLI: `list pkg`; `extract pkg outdir` (writes `outdir/files/...` and `_manifest.json`); `repack indir out.pkg [--endian le|be] [--level N]`.	Unchanged. Round trip of an untouched original is byte-identical
`wozdsc.py`	Binary property files (`.dsc`, `.pmp`, many `.xml`) and `.rlet` images: `python wozdsc.py srcFolder dstFolder`	Last changes: PlayerIDList + last-resort pass.
`wozhkx.py`	Havok `.hkx` converter: `python wozhkx.py WiiFolder PCFolder`.	Last changes: cylinder shape layout.

File	Purpose	Notes
`woztpl.py`	Wii `.tpl` to PC textures (mip chains, lossless CMPR to DXT1, PC-style TGA, placeholder textures). CLI: `python woztpl.py in.tpl --format tga|dds|both [--info]`	Changed (24-bit color-layer TGA). Its own CLI `--format tga` still writes a plain viewer-friendly TGA; the PC rules apply inside `wozconvert`.
`wozfix.py`	Optional, off-by-default Penguin-only fixes (section 12). CLI: `python wozfix.py in.hkx out.hkx hold|nursery`.	New.
`woz_langpkg.py`	Language packages (`Language*.pkg`): `info`, `extract`, `convert --to pc|wii`.	Separate format (section 9). Unchanged.
`woz_order_table.json`	Learned map-entry order and key kinds from PC files.	Optional; entry order does not matter to the game.
`woz_pkg.bms`	Obsolete QuickBMS extractor. Original non-working starting point.	No longer needed.

3.1 `wozconvert.py` options

Option	Effect
`--drop-256`	Omit the low-resolution `*_256.tpl/.rlet` variants. PC packages do not contain them; with this option the converted file list of all 11 Big Cat packages equals the real PC packages. **This is the normal way to run it.**
`--textures DIR`, `--hkx DIR`	Replacement files with the same relative paths as inside the package; they take priority over conversion.
`--dds-mode rgba|dxt1`	Default `rgba` = uncompressed 32-bit DDS with generated mips. `dxt1` = lossless CMPR repack. Both load in the PC engine.
`--no-mips`, `--keep-tpl`	No mipmaps for uncompressed DDS; keep the `.tpl` next to the converted file.
`--table`, `--ref`, `--save-table`	Optional map-entry ordering. Not needed.
`--tga-raw`	Write 24-bit color layers uncompressed instead of RLE.
`--tga-legacy`	Old behaviour: every TGA as 32-bit raw. **Animals stay gray in-game**; only for comparison.
`--penguin-hold-fix`	Optional (off by default): Penguin `Mouth_Attach_Point` pulled back 0.06 along the beak.
`--penguin-nursery-fix`	Optional (off by default): Penguin nursery ground/boundary collision bypasses the converted MOPP trees.
`--report FILE`	Also write the report to a text file.

3.2 Latest change log

Tool	Change	Why / verification
`woztpl`	Opaque color-layer TGAs are written 24-bit, RLE, bottom-left origin, with the PC footer (`is_colour_layer()` rule). Masks (`*_alpha`, `*_dirty`, anything with bodymap, sensitivity, template, dirt) stay 32-bit raw as before.	Bear stayed uniformly gray with 32-bit layers; 24-bit fixed it in game (section 8.4). The new converter reproduces the in-game-verified Bear packages byte for byte.
`wozdsc`	`PlayerIDList` added to `ARRAY_KEYS`; new last-resort retry that accepts bare counted arrays for keys never seen in PC data.	Crocodile nursery `settingmodebase.dsc`. 616 of 617 Wii property files convert identically before/after; the 617th is the fixed file. All 686 PC property files (instances across packages) still re-serialise byte-identically.
`wozhkx`	`hkpCylinderShape` word mask corrected (words 8 to 23 all swap).	Bear hammer/log bug (section 11.4). Changes exactly three files in BearForest; nothing else in any package we used.
`wozfix`	New module + two options.	Option-off output byte-identical to before; option-on equals the tested experiment files.
`wozconvert`	Re-embeds the four changed modules; new options; report lines for the optional fixes.	Re-verified end to end on Big Cat, Bear, Penguin, NurseryPool, Crocodile nursery.

4. Package inventory and naming

Package naming pattern: `[Species]Animal` (+`Preloading`, +`Editor`, +`StarAnimals`, +`Templates`), `[Habitat]` (e.g.Savanna, Pool, Forest, Paddock) with `+Icons` and`+Psys`, `Nursery[Habitat]` with `+Icons` and `+Psys`, plus `Language[Animal]` and shared packages. Entry paths look like `actors/[animal]/model/*.hkx`, `environments/[habitat]/model/*.hkx`, `textures/actors/...`, `scripts/*.dsc`. Paths inside packages are lower-case on disk; some data files reference them in mixed case, which the engine resolves case-insensitively.

Family	Packages seen in this project	Notes
Big Cat (Wii and PC)	BigCatAnimal, Preloading, Editor, StarAnimals, Templates, BigCatSavanna (+Icons, +Psys), NurseryPlaypenBigCat (+Icons, +Psys)	11 packages; ground truth for every format. The PC game ships the cat natively (the enclosure holds a maximum of 3).
Panda (native PC)	PandaAnimal, PandaAnimalPreloading, PandaAnimalEditor	Used as a second PC reference (appearance data, attachment nodes, skeleton).
Penguin (Wii)	PenguinAnimal (324 entries), Preloading (143), Editor (256), StarAnimals, Templates, PenguinPool (361), PenguinPoolIcons (62), PenguinPoolPsys (144), NurseryPool (125), NurseryPoolIcons, NurseryPoolPsys	Complete set. Only one Penguin model exists (`penguinjackass_adult_m.hkx`, in Animal and Preloading); babies are the same skeleton with a baby look applied.
Bear (Wii)	BearAnimal (326), Preloading (173), Editor (220), BearForest (470) + Icons (67) + Psys (192), NurseryBear (110) + Icons (21) + Psys (65)	`BearAnimalStarAnimals` / `BearAnimalTemplates` never provided; `LanguageBear.pkg` exists (converted earlier).
Crocodile (Wii)	CrocodileAnimal, Preloading, NurseryPaddockCrocodile (116 entries)	Nursery package converts since the `PlayerIDList` fix.
Antelope (Wii)	GazelleSavanna	Float-UV meshes (vertex type 4) supported.
Shared	`PropertyKeys.pkg`, `SharedExhibit.pkg`, `SharedAnimalEditor.pkg`, `SharedHoofedAnimal.pkg`	Section 7.4 and section 13.

5. The .pkg container [verified]

Header size = `8 + 40 * numTypes` bytes (`0x148` for 8 types, `0x170` for 9). All u32 fields are **big-endian on Wii, little-endian on PC**.

```
offset  field
0x00    u32 platform flag         Wii = 1, PC = 0   (a flag, not a byte-swapped number)
0x04    u32 0
0x08    u32 numTypes              8 normally, 9 when a "psys" type is present; READ it, never assume
0x0C    u32 numFiles              (this word is also the first field of type record 0)
0x0C    type table: numTypes records x 40 bytes = [u32 X][u32 id][char name[32]]
        type names seen: xml, texture, mat, (psys), hkxContainer, renderModel,
                         havokCharacterData, havokAnimation, PMap
then    entry table: numFiles x 288 bytes, each:
          u32 idx            0 for the first entry, 1 for all others
          u32 type           resource type id (1-based index into the type table)
          u32 ord            1-based position in the table
          char path[256]     NUL padded; bytes after the NUL are stale garbage in originals (safe to zero)
          u32 offset         LAST 4 BYTES of the 260-byte path buffer: offset of this entry's compressed data
                             = sum of the compressed sizes of all EARLIER entries   (always recompute!)
          u32 csz  u32 usz   compressed / uncompressed size
          u32 crc32(compressed block)   u32 crc32(uncompressed data)
then    8 bytes: u32 1, u32 numTypes
then    zlib streams back to back in table order (original streams: zlib level 6, reproduced byte for byte)
```

- **Wii to PC header:** swap words 4 to 12 and every record word, set the platform flag to 0. Verified identical to real PC headers.
- **The offset field matters.** Early test packages copied it unchanged (stale or byte-swapped) and the game misbehaved; a long false lead (entry ordering) followed. `wozpkg`/`wozconvert` always recompute offsets and CRCs.
- Wii packages also contain low-resolution `*_256.tpl/.rlet` variants that PC packages do not have (`--drop-256`).
- Re-packing an untouched package reproduces the original byte for byte (re-verified this session for BearAnimal and BigCatAnimal).

6. Binary property files (BFBINARY)

Files whose first 8 bytes are `BFBINARY`: all `.dsc`, `.pmp` and many `.xml` (other `.xml` are plain text and identical on both platforms). They are typed property maps that carry nearly all game data: animal families, appearance recipes, objects, controllers, building points, inventory, save records.

6.1 Layout [verified]

```
file          = "BFBINARY" + map
map           = u32 count, then count x ( key string, value )          (a map never repeats a key)
string        = u32 byteLength + ASCII bytes (no terminator)
wide string   = u32 byteLength + UTF-16 (high byte assumed 0)
vector        = u32 count, u8 flag, then
                  flag 1 (homogeneous):   ONE type-name string, then count elements of that type
                  flag 0 (heterogeneous): every element preceded by its own type-name string
values are NOT tagged. Kinds seen: bool (1 byte), u32/float (4), vector2 (8), vector3 (12), vector4 (16),
  matrix4 (64, keys *Transform), string, wide string, nested map, vector,
  counted array [u32 n][n x 4 bytes] for keys VertDifs / VertIndices / PlayerIDList (see 6.4)
vector element type names: propertymap, propertyvector, unicodestring (wide), vector2/3/4, quaternion,
  matrix4, int/float/uint, bool, IntList/FloatList (counted arrays), propertypair, stringtoken,
  enum names (strings), struct-like names (XMLRef, Resource = wide path string, PropertyMapMessageSender ...)
```

6.2 Conversion and the ambiguity problem

- Conversion swaps every u32/u16 token. Because values are untagged, the converter finds token boundaries by a memoized backtracking parse, re-serializes little-endian, and re-parses the result as a self-check.
- **Ambiguity is real and dangerous.** A 1-byte bool followed by a key can be read as the start of a vector, a nested map can look like an int followed by sibling keys, an empty string equals int 0. The self-check cannot detect a wrong-but-consistent parse.
- The fix is a **learned table key -> allowed value kinds** (750 keys from 447 PC files), built into `wozdsc.py`. Without it, exactly one of 606 Bear/Penguin files was converted wrongly (`snowmachine.dsc`) and the game crashed while parsing it.
- **Entry order is irrelevant.** The two platforms write map entries in different but per-platform consistent orders; no hash reproduced the order. The earlier belief that order mattered was caused by the stale-offset bug. Wii-order files work in game.
- 106 keys occur only in Penguin data (e.g. `TerrainType`, `PreferencesSetList`); their value kinds are unconfirmed but they converted without issue.

6.3 Validation numbers

Check	Result
PC corpus (originally 447 files: 318 .dsc, 127 .xml, 2 .pmp)	All re-serialize byte-identically. Re-checked this session on every PC package available: **686 file instances, 686 identical** (instances repeat across packages).
Big Cat Wii to PC	Exact for all 156 `.dsc` and all 291 binary files of the companion packages (with the PC-order table).
Wii property files, old vs new `wozdsc`	617 files across all Wii packages in hand: 616 identical, 1 newly parsed (`settingmodebase.dsc`).
Crocodile nursery after fix	64 binary property files converted, 0 failed.

6.4 The `settingmodebase.dsc` failure (Crocodile nursery) [verified]

The file ends with the key `PlayerIDList` whose value is a bare counted array: `04 00 00 00` followed by four u32 (0, 1, 2, 3), with no vector header. The parser only knew that layout for `VertDifs` and `VertIndices`.

- `PlayerIDList` is now in `ARRAY_KEYS`. No PC file uses it as a key, so there is no conflict.
- New **last-resort pass**: if both normal attempts fail, retry once accepting bare counted arrays for keys that never occur in the PC prior. Normal parsing is untouched. Not yet exercised beyond this one file, so check the report for files that convert via the fallback.
- Diagnostic technique used: hex-dump the failing file, walk the map structure by hand, find the first byte run that no known value kind explains ("deepest key reached").

7. Other file types and data sources

7.1 `.rlet` layer masks [verified 424/424 vs PC]

Run-length coded 8-bit images used as appearance layers and masks. Header = 11 u32 words (word 5 = header size 44, word 7 = data length; every file seen: `10, 1, 4, W, H, 44, W*H, dataLen, 2, 0, 255`; all seen files are 512x512). Stream: a `0x00` or `0xFF` byte is a run marker followed by a u32 run length; every other byte is a literal. **Conversion = swap the header words and the run lengths.** This session: all 162 converted Bear `.rlet` files decode to exactly 512x512 with header constants identical to Big Cat (212 files) and Panda (247 files).

7.2 Plain copies

`.bfmat` (XML), `.psys` (XML), `.lightrig` (XML), `.gfx`, `.jpg` and text `.xml` are identical on both platforms and are copied. Materials (`.bfmat`) name a shader (`fx`) and textures (e.g. `fx="decon_dirt"`, `diffuseMap = Textures/Actors/.../xxx_map`); converted `.bfmat` files were byte-identical to native PC for Big Cat.

7.3 The save file (`ZooInfo2`) [observed]

1,032,192 bytes. Starts with `BF01` and a small header, then `BFBINARY` property records. The file held 10 animal records. A record contains (among gameplay values): `Name` (internal, e.g. `EditorBear2`; the on-screen name is a **different** UTF-16 string, e.g. `WiiBaseTes`), `TextureColorIDList`, `AnimalPartSelectionList`, `AnimalFamily` (e.g. `Bear`), `Description` (ASCII) and `DescriptionName` (UTF-16) = `Actors/<Family>/Data/<Family>Family.dsc`, `PctDirty`, `DoBakedMorphedVertices`. In record order `AnimalFamily` comes **after** the color list, which matters when matching a record to its family.

Field	Content
`TextureColorIDList`	`vector3` list of **six RGB triplets** (18 floats, 0 to 1) = the six editor colors. The first triplet is Main Color for both Big Cat and Bear. Examples: Big Cat default (0.74309, 0.53023, 0.04644) = RGB (189, 135, 12) #BD870C; after the purple edit (0.54902, 0.18431, 0.91765) = RGB (140, 47, 234) #8C2FEA. Other five values unchanged in color-only edits.
`AnimalPartSelectionList`	Property map with one record per body part: `AnimalPartSelectionBodyPart`, `AnimalPartSelectionMorphPackage`, `AnimalPartSelectionTexturePackage`, `MorphWeight`. Texture packages look like `GRIZZLY`, `POLAR`, `BLACK`, `NAKED`, `Mask`, `K/J`; morph packages like `BaseHead!`, `Cheer!`.

- **Clean controls (re-run by the user after the fix) [verified]:** a color edit changes only the color list; a pattern edit changes only that part's texture/morph package; head morph changes only the head. Color and pattern are stored independently.
- An early Bear color-only test that appeared to rewrite many selections was an outlier from the broken (gray) state and must **not** be used to claim that color edits rewrite patterns.
- Hex pasted from the save is often truncated at the end; do not assume omitted tail fields are absent.

7.4 `PropertyKeys.pkg` and the Shared packages

- `PropertyKeys.pkg` holds one file, `properties/propertykeys.xml`: **856** `PropertyKey` / `ePropertyType` pairs. Of the keys searched inside it, only `TextureGenDesc` (`propertymap`) and `EditorPaletteColorIndex` (`int`) exist there. The searched keys, `TextureColorIDList`, `EditorTextureList`, `EditorTextureGroupList`, `EditorColorPaletteList` `AnimalPartSelectionList` are **not** in it even though saves and data use them: **the file is not an exhaustive key list.**
- `SharedExhibit.pkg` holds generic exhibit effects: about 49 particle systems including the sickness effect (section 13), plus textures and materials. Many of its psys files are never named in any data file, so the engine selects effects by name built in code.
- `SharedAnimalEditor` has only generic magic-moment scripts. `SharedHoofedAnimal` was thought to hold leftovers of unrelated hoofed animals.

8. Textures and the color / appearance system

8.1 Formats and the extension rule

- **Wii TPL formats seen:** CMPR, RGB5A3, RGBA8, I8 (decoder also implements I4, IA4, IA8, RGB565, C4, C8, C14X2, untested). Multi-image TPLs are rare.
- **PC textures:** DDS (DXT1/DXT3/DXT5, full mip chains of 8 to 11 levels) or TGA (24/32 bpp, bottom-left origin, type 2 or RLE type 10, with the `TRUEVISION-XFILE.` footer). PC textures are usually higher resolution than the Wii ones (64 to 256, 256 to 512, 4x4 stand-in to 1024).
- **Extension rule [verified, 11 of 11 packages]:** RGBA8 images and names ending in `_dirty` become `.tga`; everything else `.dds`; textures under a `particles` folder are always `.dds`.
- **Decoder checks:** RGBA8 to TGA is pixel-exact against PC. CMPR to DXT1 (`--dds-mode dxt1`) decodes to exactly the same pixels as the user's decoder and matches PC colors. The default 32-bit DDS with generated mips loads in the PC engine.
- **Stand-in textures:** a 4x4 CMPR texture (`bear_adult_m`, `decontiger_map`, `PenguinJackass`) is only a stand-in on Wii. The converter writes a **1024x1024 solid mid-gray DXT1 with 11 mips** (699,192 bytes) whose header is identical to the PC `decontiger_map.dds`. (Earlier notes of a 4096x4096 output describe an earlier iteration.)

8.2 TGA formats in native PC packages (census)

Kind	Examples	Type / bpp / descriptor
Color layers (opaque)	`bigcat_base_colorlayer_0101`, `bigcat_mane_base`	10 (RLE) / **24** / 0x0
Color layer (opaque)	`panda_basecolor_basecolor_0101`	2 (raw) / **24** / 0x0
Dirt color	`tigerbengal_adult_m_dirty`, `panda_adult_m_dirty`	2 / 24 / 0x0
Dirt alpha masks	`*_completelydirty_alpha`, `*_mostdirty_alpha`, `*_dirt_alpha`	2 or 10 / 32 / 0x8
Maps	`panda_bodymap`, `*_sensitivity`, `snagtemplate`	2 or 10 / **32** / 0x8 (even when fully opaque)

Descriptor 0x0 = bottom-left origin, no alpha; 0x8 = bottom-left origin with 8 alpha bits. Rows are stored bottom to top.

8.3 How the PC engine builds an animal's coat [observed + tested]

- **The material** (`.bfmat`) uses the shader `decon_dirt` (whiskers use plain `decon`) with `diffuseMap` = the animal's main texture. The PC game ships that texture as a full 1024x1024 DXT1 UV atlas (the Big Cat one is a flat-color painted tiger, 5,256 distinct blocks); on Wii it is a 4x4 stand-in.
- **The recipe** lives in `<family>_appearancedata.dsc` and is linked from the family file: `bearfamily.dsc` / `bigcatfamily.dsc` list `Actors/<Animal>/Data/<Animal>Family_AppearanceData.dsc` in their `ArchetypeList` (Panda's list does not, though the file exists; unexplained and harmless).
- **Structure of the recipe:** `ModelName`, `MorphData` (xml), attachment/mesh-group data, then `TextureGenDescs` = vector of property maps; each has `Name` ("Default"), `DestinationMaterials` (a vector that is **empty in Big Cat, Panda and Bear**) and `TextureGenOps`. Each op has `TextureGenOpName`, an active flag, `LayerBlenderTexture`, `LayerBlenderAlpha`, `LayerBlenderColor`, `TextureColorID` (which of the six colors) and `TextureColorModifier`.
- **Size of the recipes:** Big Cat 245 ops (217 path strings), Panda 257 ops (252), Bear 178 ops (166 path strings = 162 `.rlet`, 1 `.tga`, 1 `.hkx`, 2 `.xml`). All referenced layer files resolve in every animal (Panda has one missing editor-setting xml, irrelevant).
- **Layer files:** `.rlet` masks (512x512, 8-bit) plus one or two **opaque color-layer TGAs** (`bear_base_basecolor_0101.tga`, 512x512) that carry the base color and the mouth/nose detail. The Main Color tints the base layer; the other colors drive the pattern masks.
- **The engine code exists in the PC exe:** the key names `TextureGenDescs`, `DestinationMaterials`, `TextureGenOps`, `TextureGenOpName`, `TextureColorIDList`, `TextureColorID`, `TextureColorModifier`, `LayerBlenderTexture/Alpha/Color`, `ColorFill`, `AppearanceData` sit together in the string table of `WoZRetail.exe` (file offset `0x61f87c` onward). The consumer code was not disassembled.
- **Dirt** is a separate path (`blendColorMap` in the material) and kept working throughout, which is why it was never a clue.



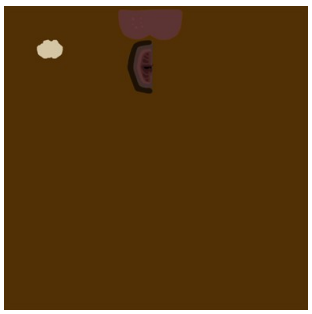
PC decontiger_map.dds (Big Cat main texture, 1024x1024): a flat-color UV atlas shipped as the material's texture.

8.4 The unpainted-animal investigation and its resolution [verified in game]

Symptom: converted Bear and Penguin rendered uniformly gray (due to the generated gray 1024x1024 that replaces Wii's 4x4); the editor accepted color and pattern changes and the save stored them correctly, but nothing changed visually.

Test	What was changed	Result	Conclusion
TEST1	Gray placeholder replaced by the Bear base-color layer as a 1024x1024 DXT1 with 11 mips.	Bear brown, **no change** with colors or patterns; mouth texture correct.	Engine shows whatever the file contains; the generator is not taking over the Bear's material.
TEST2 (control)	Native PC Big Cat with its `decontiger_map.dds` replaced by the gray placeholder.	Big Cat **still colored** and recolored correctly.	Generation does not depend on the shipped texture. The fault is specific to the converted data.
TEST3	Bear `bear_base_basecolor_0101.tga` re-encoded in PC style: 24-bit, RLE, bottom-left, PC footer (pixels identical).	Bear **fully works**: colors and patterns change as expected.	**Root cause: the converter wrote opaque color layers as 32-bit raw TGAs; the generator needs the PC form.**

- **Not individually separated:** whether the 24-bit depth, the RLE, or the descriptor is what matters. Native Panda ships 24-bit **raw** layers, so both 24-bit forms are native; the 32-bit raw form is the one that failed. The tools write 24-bit RLE (the verified form); `--tga-raw` writes 24-bit raw.
- **Ruled out along the way:** missing layer files (all resolve), broken `.rlet` (all decode), different descriptor layout (Bear, Panda and Big Cat match), missing family link, wrong key types in the property files, the stand-in texture size.
- **Not touched on purpose:** masks, dirt maps, `bodymap`, `sensitivity` and templates stay 32-bit raw (PC keeps `bodymap`/`sensitivity` 32-bit even when opaque; changing them is untested).
- Big Cat with the new converter: the two layer TGAs match native PC (`bigcat_mane_base.tga` byte-identical, `bigcat_base_colorlayer_0101.tga` pixel-identical, one byte of run layout apart).



bear_base_basecolor_0101.tga (512x512): the Bear base color layer - brown with mouth and nose detail; the full coat is composed from this plus 162 masks.

9. Language packages (`Language*.pkg`) [verified on LanguageBear (Wii) and LanguagePanda (PC)]

A different, smaller container handled only by `woz_langpkg.py`. All ints use the file's endianness.

```
0x00 u32 flag (Wii sample 1, PC sample 0; meaning not confirmed)   0x04 u32 zero
0x08 u32 nTypes (2: "raw", "PMap")                                 0x0C u32 nEntries (4)
0x10 nTypes   x 0x28 : { u32 id; char name[32]; u32 one }
     nEntries x 0x120: { u32 index(1..n); char name[0x100]; u32 dataOffset(from end of tables); u32 compSize;
                         u32 rawSize; u32 crc32(compressed); u32 crc32(raw); u32 one; u32 one_or_two }
     then the zlib streams back to back in entry order (level 6 reproduces the PC blobs byte for byte)
```

- `*.sbin`: u16 BOM (FEFF = BE, FFFE = LE), u16 count, count x { char key[20]; u32 offset (in u16 units) }, then the text blob: per string { u16 nChars; UTF-16 chars }. Conversion swaps every u16 and u32 and validates that offsets stay consistent.
- `*.pmp`: `BFBINARY` PMap whose payload is a `GUILocalizedTextFileList` (list of UTF-16 names).
- Anything else is byte-copied and reported, never guessed. The `flag` value for the target platform is an assumption taken from one Wii and one PC sample. The user reported that language packages extract fine.

10. Havok `.hkx` files [verified unless noted]

10.1 Packfile structure

- Havok **6.0.0-r1** packfiles, magic `0x57E0E057 0x10C0C010`. Layout rules at header bytes `0x10..0x13`: **Wii** (pointer 4, big-endian, reusePadding 1, emptyBase 1) vs **PC** (4, little-endian, 0, 1). Header `0x40` bytes, then three section headers of `0x30` bytes: `__classnames__`, `__types__`, `__data__`. Each section header: name[19] + `0xFF`, then 7 u32: absolute start, local, global, virtual, exports, imports, end (offsets relative to the section start).
- **Class-name entries** are `[u32 signature][0x09][name\0]`. **Fixup tables** at the end of the data section: **local** (src, dst; 8 bytes) for same-section pointers, **global** (src, section, dst; 12 bytes), **virtual** (object offset, section, class-name offset; 12 bytes). Each ends with `0xFFFFFFFF` padding; the local table is padded to a multiple of 16 bytes.
- **Important detail found this session:** pointers to objects in the *same* section can sit in the **global** table too. Skeleton bone-pointer arrays and rigid-body shape pointers are global-table entries. Any tool that follows pointers must read **both** tables.
- Object offsets in the virtual table are relative to the data section start. Re-laying out the data section (the mesh conversion does) means remapping all three tables, the section header and the object size fields.

10.2 Conversion principle

- The `__types__` section embeds `hkClass` definitions (48 bytes: name@0, parent@4 via a GLOBAL fixup, objectSize@8, interfaces@12, enums ptr@16 count@20, members ptr@24 count@28, defaults@32, attributes@36, flags@40, describedVersion@44). `hkClassMember` is 24 bytes (name, class, enum, type u8@12, subtype u8@13, cArraySize u16@14, flags u16@16, offset u16@18, attributes@20). `hkClassEnum` is 20 bytes inline, items 8 bytes. Member type numbers follow the Havok enum (BOOL 1 ... REAL 11, VECTOR4 12 ... POINTER 20, ARRAY 22, ENUM 24, STRUCT 25, SIMPLEARRAY 26, HOMOGENEOUSARRAY 27, VARIANT 28, CSTRING 29, ULONG 30, FLAGS 31).
- The converter walks every object (virtual fixups), swaps each member by type, follows array pointers (local fixups) and swaps elements, swaps class-name signatures, all fixup tables and headers, and sets the layout bytes to (4,1,0,1).
- **Class layouts are identical on Wii and PC** (133 common classes compared), so conversion is a pure byte-order job. Animation files have an empty `__types__`, so `wozhkx` carries a **pool of 146 class definitions** taken from real files.
- **Spline-compressed animation data** (`hkaSplineCompressedAnimation.data`) is little-endian on both platforms: copied unswapped. Wii keeps its spline/delta animations; the PC build loads spline files too (115 PC animations are uncompressed instead).
- `hkbStateMachineStateInfo.inPackfile` is 1 on Wii and 0 on PC: written as 0. Remaining differences from PC exports are float noise and stale runtime addresses inside mesh headers (harmless).

10.3 Hand-learned layouts (`PHYS_MASKS`, 29 classes)

For classes with no embedded definition the layout was learned by comparing aligned objects in Wii/PC pairs. Mask per 4-byte word: `4` = swap u32, `2` = swap two u16, `a` = swap only the first 2 bytes (a u16 followed by two single bytes; getting this wrong caused the original physics hang), `.` = unchanged. Arrays hanging off these objects use block rules (width swap or per-element patterns). Classes in the table include: `hkpConvexTranslateShape`, `hkpSphereShape`, `hkpBoxShape`, `hkpConvexTransformShape`, `hkpListShape`, `hkpCapsuleShape`, `hkpConvexVerticesShape`, `hkpMoppBvTreeShape`, `hkpMoppCode` (bytes unchanged), `hkpCylinderShape`, `hkpSimpleMeshShape`, `hkaRagdollInstance`, `hkaSkeletonMapper` (simple mappings start with two u16), `hkpRagdollConstraintData`, `hkpLimitedHingeConstraintData`,

`hkpStiffSpringConstraintData`, `hkpPositionConstraintMotor`, `hkpBallAndSocketConstraintData` (atoms at words 4 and 16, translations at words 8 to 15), `hkpPhysicsSystem`, `hkpPhysicsData`, `hkxAnimatedFloat/Vector/Quaternion/Matrix`, `hkxSparselyAnimatedBool/Int/String`, `hkxMaterialEffect`, `bfSubMaterialIndex` (u16 + padding, no base class). Inline `hkClass` objects inside `__data__` and `HOMOGENEOUSARRAY` members (`hkxVertexBuffer`) are supported.

Lesson from the cylinder bug: `hkpCylinderShape` had been "learned" from data where the cylinder axis was symmetric, so the x components of both end points were not marked for swapping. Old mask `....444..444.444.44.4.4.`; correct mask `....444.4444444444444444` (words 4 to 6 floats, word 7 padding, words 8 to 23 four full 4-float vectors). A mask learned only from symmetric values (often zeros) can silently hide a missing swap.

Classes in the table that never occurred in the Wii data examined (so their masks are unvalidated by value checks): `hkpStiffSpringConstraintData`, `hkpBallAndSocketConstraintData`, `hkpSimpleMeshShape`.

10.4 Blue Fang mesh buffer: `bfFastMeshBuffer` (animal and prop geometry)

```
object (16 bytes): vertexData ptr + byte size, indices ptr + count (u16 indices, plain swap)
vertex blob header: Wii 288 bytes, PC 272 bytes. Words: [0]=total bytes [1]=nverts [2]=nverts,
  then 9 slot records of [offset, stride, comps, type, flag, extra4bytes]; word 57 = distance from
  byte 228 to the data start (60 Wii / 44 PC); words 58..66 = stale runtime pointers (garbage, harmless);
  a 0xDDDDDDDD filler word.   Slots: 0 position, 1 normal, 6 (8 floats), 7 vertex colour word, 8 texcoords
```

- **Layout A "float stream"** (animal bodies; slots 0, 1, 6, 8): position and normal streams grow from 3 to 4 floats (w = 0.0).
- **Layout B "packed interleaved"** (props, habitats, editor models; slots {0,1,8} or {0,1,7,8}): fixed-point components. `extra bytes = [fractional bits, kind]`; kind 4 = u8, 5 = u16, 6 = s8, 7 = s16 (UVs unsigned, positions/normals signed); value = integer / 2^fractionalBits. `type 4` means a plain big-endian float (gazelle UVs). Slot 7 is a packed RGBA color word: bytes reordered to BGRA. PC result: interleaved float vertices of 32 or 36 bytes.
- Because buffers change size, the data section is re-laid out: blocks stay 16-byte aligned, every fixup is remapped, section header fields and the object size field are updated. The converted Big Cat adult model matches the real PC model in size, all three fixup tables and 3,704 of 3,797 data blocks; the rest is noise.
- Packed meshes match PC to within one fixed-point step (the Wii data is lower precision); two editor meshes wrap positions that exceed the representable range in the Wii data itself. Converted props/habitats therefore use slightly coarser vertices than native PC.

10.5 Object layouts measured in this work (offsets in bytes from the object start)

Class	Layout (PC little-endian; the Wii file has the same offsets)
`hkpBoxShape`	convex radius @0x10; `halfExtents` vec4 @0x20 (x, y, z, w; w equals the smallest half extent in every box seen). Object size 0x30.
`hkpCylinderShape`	24 words: word 4 convex radius, word 5 cylinder radius, word 6 base-radius factor, word 7 padding, words 8 to 11 vertexA (w = cylinder radius), 12 to 15 vertexB, 16 to 19 perpendicular1, 20 to 23 perpendicular2.
`hkpConvexVerticesShape`	aabb half extents @0x20, aabb centre @0x30; `rotatedVertices` array fixup @64 (n blocks; each block = 12 floats: x[4], y[4], z[4]); `numVertices` @0x4C; `planeEquations` array fixup @80 (planes n.x, n.y, n.z, d with n.p + d = 0 on the surface; a slab has 6). Size 288 for an 8-vertex hull.
`hkpConvexTransformShape`	child shape pointer fixup @24; transform @words 8 to 23: three **column** vectors (words 8 to 11, 12 to 15, 16 to 19) then translation (words 20 to 23).
`hkpConvexTranslateShape`	translation @0x20.
`hkpMoppBvTreeShape`	fixups: +20 to the `hkpMoppCode`, +52 to the child shape (a `hkpListShape`).
`hkpRigidBody`	shape pointer is a (global-table) fixup; in the nursery the placement transform is identity. Name string reachable through a fixup inside the object (`Proxy_RB_Ground`, ...).
`hkxNode` (scene node)	name pointer @+8; keyframe array pointer @+20, count @+24 (each keyframe = 16 floats, **row-vector 4x4**: rotation rows = transpose of the skeleton's column-convention rotation, translation = row 4); children array pointer @+28, count @+32, capacityAndFlags @+36. Offsets measured empirically in these files.
`hkaSkeleton`	fixup order: name, `parentIndices` (int16 array), `bones` (array of 4-byte pointers, **global-table fixups**; each bone name is a pointer to a C string), `referencePose` (hkQsTransform = translation[4], rotation quaternion xyzw [4], scale[4] = 48 bytes), `referenceFloats`.
`hkaSkeletonMapper`	simple mappings: stride 64 {u16 boneA (ragdoll), u16 boneB (animation), padding, hkQsTransform @+16}; chain mappings: stride 112 {4 x u16 startA, endA, startB, endB, padding, start transform @+16, end transform}; unmapped-bones u16 array; extracted-motion transform @object+0x40.
`hkaDefaultAnimatedReferenceFrame`	`up` @0x10, `forward` @0x20, `duration` float @0x30, sample array pointer @0x34, count @0x38; each sample vector4 (x, y, z, w).

10.6 Audit methods used

- **Byte equality against PC** wherever a PC counterpart exists (Big Cat). Accept only explicit, narrow noise classes (float noise; stale addresses).
- **Word-by-word class comparison** Wii vs converted: each word must be a plain 4-byte reversal, identical, or an expected `2/a` pattern. Run on the whole nursery: no anomaly.
- **Value sanity scan** across 755 Wii `.hkx` files: for every class in `PHYS_MASKS`, list words left unswapped (`.`) that are non-zero and look like a sensible non-integer float when read big-endian. After the cylinder fix the result is empty.
- **Geometry checks:** convex hull vertex arrays equal Wii values; skeleton poses, scene-node matrices, mapper transforms bit-identical (Penguin).
- A cruder "words unswapped but float-like" scan over arbitrary objects is too noisy (pointers, strings); only the class-aware version is reliable.

11. Models, physics and gameplay data

11.1 Skeletons, scene nodes, attachment points, ragdoll

- Each animal model holds two skeletons: the animation skeleton (Penguin: 68 bones, root `Pelvis`) and the **ragdoll** skeleton (Penguin: 13 bones - Pelvis, Chest, LLeg1, RLeg1, RArm1, LArm1, LLegAnkle, Neck1, RLegAnkle, LArm2, RArm2, Head, BeakUpper; there is **no** `Ragdoll_BeakLower`). A `hkaSkeletonMapper` links them (Penguin: 13 simple mappings, 7 chain mappings, 44 unmapped bones; the beak chain maps ragdoll Head to BeakUpper onto animation bones 14 to 23). All mapper transforms were verified bit-identical after conversion.
- **Held items attach to a scene node named** `Mouth_Attach_Point`, not to a skeleton bone. The skeletons only contain obsolete points (`Mouth_Attach_PointOLD` in Bear/Panda, `JUNKOLDMouth_Attach_Point` in Penguin, `LHand_Attach_Point`). The node is a **child of a ragdoll bone node**: `Ragdoll_Head` in Bear, Panda and Big Cat, but `Ragdoll_BeakUpper` in the Penguin (whose ragdoll has a separately jointed beak).
- Penguin node: local translation (0.1161, 0.0009, -0.0248); in the rest pose it sits at model-space (0, 0.756, 0.211), about 5 cm beyond the end of the lower beak (the adult beak tip). The node's local +x axis points along the beak. The node has two identical keyframes.
- Penguin family settings read from `penguinfamily.dsc`: `ActorCollisionScaleFactor` 2.0 (Bear and Big Cat 1.0), `babyAnimalHeadScale` 1.1 (Bear 1.3, Bear `babyAnimalEarScale` 1.3). The Penguin family file also contains `AN_SYMPTOM_*` keys (not analysed).
- All Penguin attach data (120 scene nodes, both skeleton poses, all mapper transforms) converts **bit-identically** from Wii.

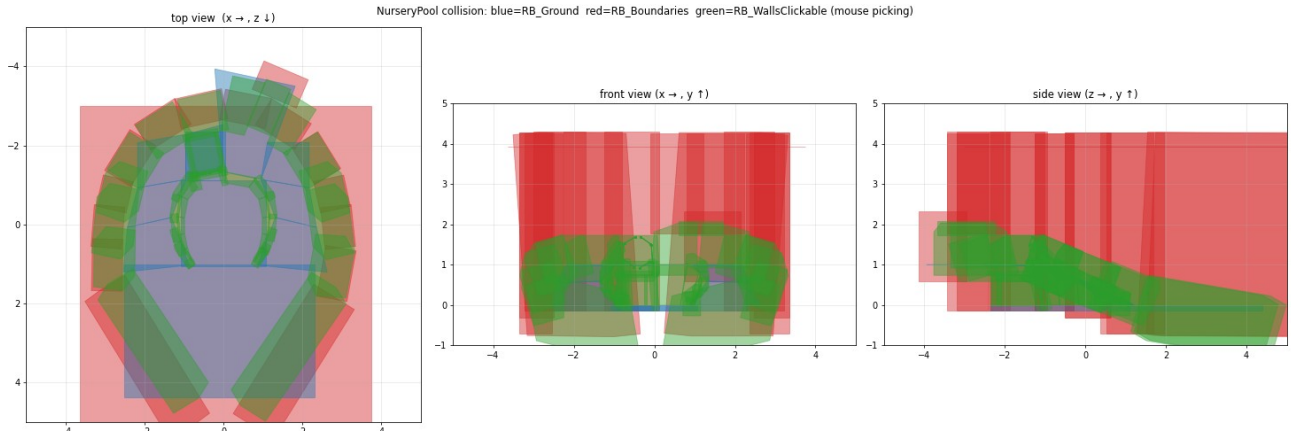
11.2 Animations and behaviour

- `bear.xml` (like `bigcat.xml`, `panda.xml`) lists animations by file name; many are not shipped on PC: Bear lists 35 absent entries, Big Cat 245 of 461, Panda 28. A long list of unused entries is normal and not a defect.
- Bear animations contain `hkSimpleLocalFrame` objects (identity transforms; Panda has them too). Root motion is intact after conversion: `Bear_MagicMoment_Attack` has a 3.07 s duration, 93 reference-frame samples and a **7.07 m forward lunge**, identical in the Wii and converted files, no garbage values.
- Behaviour graph states are the `zz_*.hkx` files (e.g. `zz_standattention`, `zz_investigate`, `zz_trot`); event names for the magic-moment attack match Big Cat. The attack controllers (`basicmagicmomentattack*.dsc`) are generic and the same size for Bear and Big Cat.
- All five Bear attack animations are present in `BearAnimal`. The in-game attack failed to start on the first run, despite there being no missing files, lost root motion, or different event/controller names (section 13). Subsequent runs worked normally.

11.3 The Penguin nursery collision (`NurseryPool`)

Everything physical in `environments/nurserypool/model/nurserypool.hkx` converts as plain byte swaps (boxes, convex hulls, list shapes, MOPP trees, rigid bodies; convex vertex arrays equal Wii values). Structure:

- `Proxy_RB_Ground`: MOPP tree over a list of convex hulls and boxes. Ground boxes are about 14 cm thick; the slope pieces are hulls with half extents of about (0.88, 0.29, 0.69); **four hull slabs only 1 cm thick** form the platform at y 0.99 to 1.0 at the back (centres (-0.476, 0.995, -1.832), (0.800, 0.995, -2.535), (1.564, 0.995, -1.502), (-1.564, 0.995, -1.602)).
- `Proxy_RB_Boundaries`: MOPP tree over a list of boxes. The walls are thick (half extent about 0.39, so about 0.8 m); one box is a **ceiling plane** of 7.39 x 8.36 m at y of about 3.92 with essentially zero thickness (z half extent 6e-7).
- `Proxy_RB_WallsClickable` is the mouse-picking set (left untouched), and further `Proxy_*` bodies (food/toys areas) exist. All rigid bodies sit at the identity transform.
- The data layout is the same as the Wii's. The converted **MOPP trees** are the suspect: bypassing them (N1) fixed the glitches in game. Whether the MOPP bytecode is corrupt or merely not valid for the PC engine is **unverified**; regenerating MOPP code needs Havok's own builder, which is not available.



Nursery collision pieces drawn in three views (blue = RB_Ground, red = RB_Boundaries, green = RB_WallsClickable). The thin horizontal line near y 3.9 is the ceiling plane; the 1 cm slabs are at y 1.0.

11.4 The Bear hammer/log bug (cause and fix) [verified in game]

- The hammer swaps items that share a **building point**. `BF_Center_PlaysetRootC` defines five points: three allow {Beehive, Log, Tree}, one {Beehive, TreeMirrored}, one {Beehive, Log, TreeMirrored}. Objects are `bf_center_beehive.dsc`, `...beehivelog.dsc`, `...beehivetree.dsc`, `...beehivetreemirrored.dsc` (scripts `InExhibitBuildItem`; keys `BuildingTags`, `BuildingPoint`, `AllowedTags`, `DefaultBuildItemDescriptionSet`, `CanCycleToNothing`).
- **Symptom:** once a beehive became a log it could not be swapped again; only the slot with no log option kept working; the gazelle enclosure, used as comparison, was fine.
- **Cause:** the log model is the only one of the four using a `hkpCylinderShape` (plus `meatbarrelfeeder`, `toylog`, `waterwheel`), and the learned cylinder layout missed the x components of both end points (converted value about -9.6e31 instead of -0.0541). A collision shape that broken cannot be clicked.
- **Fix:** corrected mask in `wozhkx`; reconverting changes exactly `bf_center_beehivelog.hkx`, `toylog.hkx` and `waterwheel.hkx`. Result confirmed working in game. `PenguinPool` has two cylinders (fish food models) with x = 0, so it was unaffected; the NurseryPool packages contain none.

12. Optional Penguin-only fixes (off by default)

These are **workarounds for engine-side differences**, not corrections of conversion errors (the converted data equals the Wii data). They live in `wozfix.py`, each patches exactly one specific file, and each refuses to run if the file does not match.

Option	File patched	What it does	Test evidence
`--penguin-hold-fix`	`actors/penguinjackass/model/penguinjackass_adult_m.hkx` (in PenguinAnimal and PenguinAnimalPreloading)	Moves both keyframes of `Mouth_Attach_Point` back along the beak by 0.06 model units (x 0.1161 to 0.0561). Refuses to run twice.	**Test B chosen:** Held items sit closer to the beak on babies and adults. **Does not** remove the wobble.
`--penguin-nursery-fix`	`environments/nurserypool/model/nurserypool.hkx` (in NurseryPool)	The rigid bodies `Proxy_RB_Ground` and `Proxy_RB_Boundaries` point at their list shapes instead of their MOPP trees (two pointers in the fixup table). Nothing else changes.	**Test N1 chosen:** Nursery collisions (ball through wall, slope and corner glitches) work better.

Experiments that were tested and not adopted

- **A** (pull-back 0.03): built, not chosen. **B** (0.06): adopted.
- **C** (re-parent `Mouth_Attach_Point` under `Ragdoll_Head` with a compensated transform and the 0.06 pull-back): items passed through the Penguin's head; **discarded**.
- **N2** (N1 plus the four 1 cm slabs extended down by 10 cm and the ceiling given 0.2 m thickness): built, not needed because N1 alone worked; not in the tools.

13. Known bugs, open questions and next steps

Issue	What is known	Ruled out	Needed / next
Bear attack animation: the cutscene sequence plays but the bear stays in place. Other Bear actions animate correctly.	Animation, controller and event names exist and match Big Cat; root motion survived conversion exactly.	Missing animation files; lost root motion; different event or controller names.	Find out if something in Havok Engine leads to animations or particles failing / missing on first load. **Non-issue.**
Missing green sickness bubbles on a sick Bear (the scanner minigame works).	`SharedExhibit.pkg` has the effect `animalinteraction_ai_magicmomentrequest_sick.psys` (green emitter, material `particle_magicmoment_sickbubble`, texture `particle_bubble`); the bubble wand toy uses the same bubble texture. Nothing in any data file names the effect, and the exe does not contain the name as a string, so the engine builds or selects it by a rule not visible in the data. Bear has `Bear_Walk_Sick` and `Bear_Stand_2Sit_Sick`.	A missing Bear reference in the family, attachments or shared data that we can see.	Turned out same as above. **Non-issue.**
Penguin held items wobble items in the nursery are held slightly lower than the cursor.	Hold point B improves position. Attach point sits under the beak's own ragdoll body in the Penguin only. Penguin family has `ActorCollisionScaleFactor` 2.0. All attach data converts exactly. Crocodile babies hold items exactly like Wii; native PC animals hold slightly better.	Conversion error in model/skeleton/mapper data; re-parenting to the head (C).	User suspects the held items themselves (too wobbly, or a displaced hitbox). Possible check: compare the rattle, fish and chum rigid bodies and their constraints against items that behave well.
Nursery MOPP trees convert badly for the PC engine.	Bypass N1 works. Bear forest (26 MOPP trees) and PenguinPool (22) also contain MOPP trees and show no reported problem.	Geometry conversion errors.	If any other converted habitat shows leaking collision, the same bypass applies; it could be generalized Root cause of the MOPP problem is unverified.
Crocodile repeated item requests.	Original Wii behavior (user confirmed).	-	None; no fix needed. **Non-issue.**
Unvalidated converter layouts.	`hkpStiffSpringConstraintData`, `hkpBallAndSocketConstraintData`, `hkpSimpleMeshShape` never occurred in the data checked; 106 Penguin-only property keys have unconfirmed value kinds.	-	Validate when such data appears.
Cosmetic differences vs native PC.	Packed-vertex meshes are coarser (Wii precision); Bear family lacks an `EditorSetting` key that Panda and Big Cat have; Panda's `ArchetypeList` omits its appearance data although the file exists.	-	Harmless so far.
Extra modding tools and their documentation	Script for adding colors (`woz_palettemod.py`) and expanding exhibit slots to 6.	-	Planned to be included later.

14. Corrections to earlier documents

Earlier statement (source)	Correct statement
"`EditorTextureList`, `EditorTextureGroupList`, `EditorColorPaletteList` are defined in `PropertyKeys.pkg`"	Only `TextureGenDesc` and `EditorPaletteColorIndex` exist there; the other names are not defined in it and `PropertyKeys.pkg` is not exhaustive.
`.rlet` conversion may be "syntactically converted but semantically wrong"	Verified exact against PC (424/424), and every converted Bear layer decodes to a full 512x512 image.
"Converted file inflated to 4096x4096" vs "1024x1024 placeholder"	Two iterations. The current placeholder is 1024x1024 DXT1 with 11 mips.
Previous `.hkx` statements, "5 physics classes / 17 learned layouts", and early crash status	Superseded. See section 10: 29 learned layouts, 146 pooled class definitions, model and enclosure conversion complete.
"The PC engine does not composite the layers by itself; either understand it or bake a texture ourselves" (earlier summary)	The engine does composite; the converted 32-bit layer TGAs were the cause. No baking is needed.
Converter writes every RGBA8 texture as 32-bit TGA (earlier summary)	Opaque color layers are 24-bit RLE now; masks and maps unchanged (section 8.4).

Earlier statement (source)	Correct statement
`scripts/settingmodebase.dsc` "could not parse" (earlier summary, open issue 1)	Fixed (section 6.4).
`hkpCylinderShape` layout "learned from Wii/PC pairs" (earlier summary)	It was wrong in five words and never validated against a PC cylinder; fixed (section 10.3).
Hypothesis: the Bear's hammer failure might be a misparsed property value	Ruled out; the keys involved are PC-confirmed. Cause was the cylinder collision shape.
Hypothesis: nursery walls are paper-thin	Walls are thick. The thin pieces are the ceiling plane and four 1 cm slabs. The fix that worked is the MOPP bypass.
`Mouth_Attach_Point` is a skeleton bone (assumed)	It is a scene node under a ragdoll bone node; the skeleton only holds obsolete `...OLD` points.

15. In-game test log (chronological)

15.1 Earlier work: packages, property files, Havok, Penguin set

Test	Result	Lesson
Wii pkg + converted `.dsc` (Test 1)	Area loads, Big Cats invisible.	Models and animations still in Wii format.
+ PC `.dds` textures (Test 2)	Crash.	`.hkx` files unconverted.
PC pkg + all converted `.dsc`, halves A and B	A: no cats; B: cats with wrong behavior	Looked like entry order; truly the stale data offsets.
PC pkg + Wii-order `.dsc`, fixed offsets	Works normally.	Offset field was the cause; order irrelevant.
PC pkg + Wii-converted `.hkx` except models	Works perfectly.	Behavior, character, animation conversion correct.
Preloading model tests 0 to 3	Control OK; mesh-only OK; scene-only OK; physics-only HANGS.	Physics layouts wrong (`a` mask words).
Preloading tests 4 and 5 (fixed)	Both work; full converted cat model loads.	Physics fix good.
Savanna + NurseryPlaypenBigCat with Wii `.hkx`	Both load, no new errors.	Packed vertex conversion correct for habitats and props.
Full Penguin set	Loads; nursery, toys, babies fine; pool enclosure CRASHED.	Crash dump analyzed (section 16.3).
After snowmachine fix + placeholder textures	Enclosures work, penguins interactive; bodies gray; dirt works.	Textures still open at that point.

15.2 Last tests done

Test	Result	Conclusion
TEST1 (Bear, base layer as main texture)	Brown, not recolorable; mouth correct.	Generator not active for the Bear material.
TEST2 (Big Cat with gray placeholder)	Still colored and recolorable.	Shipped texture irrelevant to generation.
TEST3 (24-bit RLE color layer)	Bear fully works.	Root cause found (section 8.4).
Save-file controls repeated after the fix	Color, pattern, head texture and head morph edits behave independently.	Editor and save pipeline fine.
Crocodile nursery converted with the new `wozdsc`	Converts; all animals then work.	Section 6.4.
`BearForest.pkg` with the cylinder fix	Hammer swapping on logs works.	Section 11.4.
Penguin hold point A/B (0.03 / 0.06)	B (0.06) better for babies and adults; wobble remains.	Adopted as `--penguin-hold-fix`.
Penguin hold point C (head parent)	Items pass through the head.	Discarded.
Nursery N1 (MOPP bypass)	Collisions better.	Adopted as `--penguin-nursery-fix`.
Nursery N2 (N1 + thickness)	Not needed.	Not adopted.

16. Procedures, diagnostics and lessons

16.1 Checklist for converting a new Wii animal

- Collect all its packages in one folder (Animal, Preloading, Editor, StarAnimals, Templates, habitat + Icons + Psys, nursery + Icons + Psys, language pack).
- Run `python wozconvert.py WiiFolder PCFolder --drop-256` and read the report. For the Penguin, add `--penguin-hold-fix` and `--penguin-nursery-fix` if wanted.
- "class definition missing: X" for an `.hkx`: learn the layout of X from aligned Wii/PC objects (or the file's own data) and add a mask to `PHYS_MASKS`; **then check the values** of any new layout (section 10.6).
- "could not parse" for a property file: find the deepest key reached and the bytes there, add the missing value kind or array key to `wozdsc`, re-validate the PC corpus.
- "bfFastMeshBuffer: component kind/type": a new vertex component encoding; compare against PC float values like the earlier layouts.
- After conversion in game: check colors **and patterns in the editor** (a gray animal means color layers were not 24-bit or something else broke generation), the hammer/swap tool in the habitat, nursery collisions, held items, sickness and attack animations.

16.2 Techniques used (re-usable)

- **Fixup-aware pointer following** for any Havok object (read local and global tables, see 10.1); object extents = distance to the next virtual-table object.
- **Controlled comparison:** when a native PC counterpart exists, change one thing in it (TEST2) to learn whether the engine depends on that thing.
- **Bypassing a suspect structure instead of fixing it** (N1) when the faithful fix is impossible (MOPP code cannot be regenerated).
- **Rendering the data** (the nursery collision picture) to see thin or missing geometry that tables of numbers hide.

16.3 Crash dump analysis (method that found the `snowmachine.dsc` bug)

(Windows Error Reporting, dump style 2, was activated through registry to deal with the game's removed crash dumper). `WoZRetail_exe_*.dmp` are full-memory minidumps (hundreds of MB). A Python script parsed the header, stream directory, the exception stream (code 0xC0000005, write at 0x31da8c, fault EIP 0x50be7d in `WoZRetail.exe`, image base 0x400000), the faulting thread's x86 CONTEXT (registers start 4+24+112 bytes into the stream), and the Memory64List to read memory; it walked the stack for return addresses into the exe and for ASCII/UTF-16 strings. The stack held property keys (`AttachmentDescriptionList`, `DeactivateEvents`) that identified the file being parsed (`animatedobjects/snowmachine/data/snowmachine.dsc`). The routine sets a string length and writes the terminator at `buffer[length]`; the length was garbage from two mis-swapped bytes.

16.4 Lessons that should guide further work

- Verify with the PC counterpart whenever one exists; byte equality is the strongest test; accept only explicit, narrow noise.
- A checker that is too forgiving hides real bugs (integer fields once classified as float noise caused the physics hang).
- Layouts learned from instances cannot distinguish alternatives when values are symmetric (cylinder bug): prefer byte-permutation analysis and known structure, and sanity-check with real values.
- Self-consistency checks cannot catch a wrong-but-consistent parse; use external knowledge (the PC-learned key kinds).
- Always recompute container offsets and CRCs; never copy stale bytes.
- The engine can be stricter than the file format: a pixel-identical 32-bit TGA was rejected where the 24-bit one worked. Match the **native form** of a file, not only its content.
- Look further into Wii behavior and engine-side differences when the converted data equals the Wii data bit for bit; think of workarounds to PC-related errors whenever possible while the SDK or Engine itself is unavailable to use.

Appendix A. Quick reference numbers

Item	Value
PC property files re-serialised identically (instances)	686 of 686 (originally 447 files, 750 learned keys)
Wii property files, old vs new `wozdsc`	617 files, 616 identical, 1 newly parsed
`.rlet` exact vs PC	424 of 424; all 162 converted Bear layers decode to 512x512
Wii `.hkx` converted without error	908 of 908 (+ Crocodile 117, Gazelle 114)
Wii `.hkx` scanned for layout value sanity	755 files
Class definitions pooled in `wozhkx` / common Wii-PC classes	146 / 133
Hand-learned layouts (`PHYS_MASKS`)	29 classes
Big Cat packages whose converted file list equals PC (`--drop-256`)	11 of 11
Big Cat converted model vs PC	same size, all fixup tables; 3,704 of 3,797 data blocks identical
zlib level of original packages	6
PC main texture placeholder	1024x1024 DXT1, 11 mips, 699,192 bytes
Appearance recipe sizes	Bear 178 ops, Big Cat 245, Panda 257
`PropertyKeys.pkg`	856 keys
Penguin skeleton / ragdoll bones; mapper	68 / 13; 13 simple + 7 chain mappings
Nursery hull slabs / ceiling	4 slabs of 1 cm at y 1.0; ceiling plane 7.39 x 8.36 m at y 3.92

Appendix B. Early history that still matters

- Early on, the user tested `wozpkg.py` and created a QuickBMS (checksums OK) and tried to a repackage a Wii package with `.tpl` swapped for PC `.dds` loads in the PC game but crashed once used; the first converted `.dsc` made Big Cats invisible; the offset-field bug was found by bisecting packages.
- Crash dumps obtained: `WoZRetail_exe_2400.dmp` (entering the Bear enclosure with partially converted data), `WoZRetail_exe_10756.dmp` (after testing a converted `LanguageBear.pkg`), `WoZRetail_exe_7992.dmp` (after replacing all bear files by PC Panda files while keeping the converted `LanguageBear.pkg`; the game then got further).
  These predate the `.hkx` and vertex converters.
- The user uploaded `WoZRetail.exe` and `PatchCFG.exe` earlier while looking for logging or debugging facilities; no debugger setup was needed. Suggested non-official Havok tools or an old available SDK version of the Havok Engine was not needed.
- **Open working agreement:** the user decided which optional fixes enter the tools; tests are one change at a time; the user reported in-game behavior; unresolved items have been listed in section 13 as the user confirmed them fixed or not.